## EXERCICE 1

We define the sequence $a$ by

$$\begin{cases} a_0 = 200, \\ a_{n+1} = 0.75a_n + 5, \text{ for all natural numbers } n. \end{cases}$$

Copy and complete this algorithm so that it outputs the index at which the sequence $a$ becomes strictly less than 20.001 :

```
a = .....
n = .....

while a ..... :
    a = .....
    n = .....

print(.....)
```